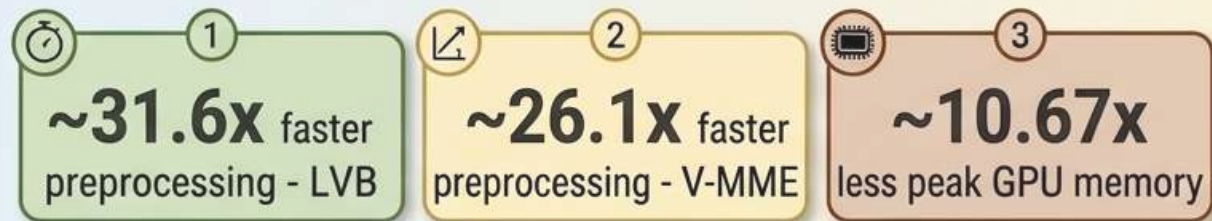


(a) High-Level Efficiency & Tradeoff Summary

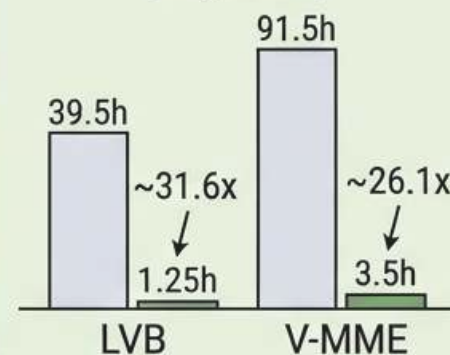


Tradeoff Analysis: $\frac{31.6\times (\text{avg speed gain})}{\sim 5\% (\text{avg relative accuracy drop})} \rightarrow \sim 6\times$ efficiency multiplier

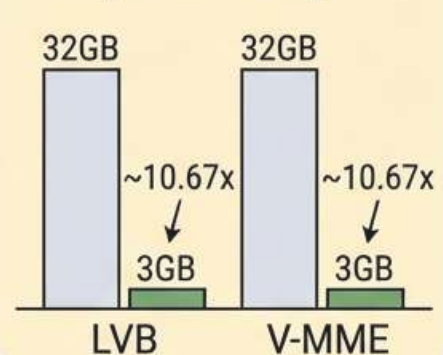
For every 1% *relative* accuracy traded, CATS delivers $\sim 6\times$ speedup.

(b) Detailed Preprocessing Efficiency Metrics

Preprocessing Time (Hours)
[Log-Scale]

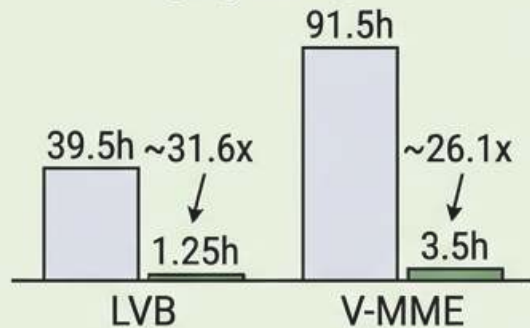


Peak GPU Memory (GB)
[Linear-Scale]

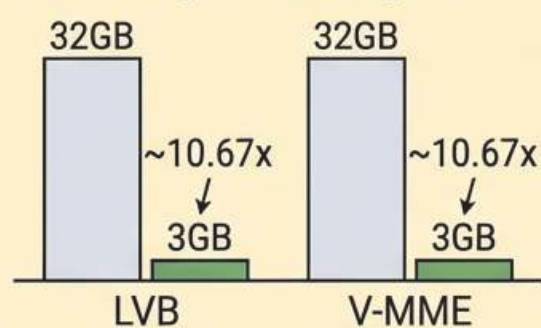


(c) Detailed Preprocessing Efficiency Metrics

Preprocessing Time (Hours)
[Log-Scale]



Peak GPU Memory (GB)
[Linear-Scale]



CATS (ours) efficiency

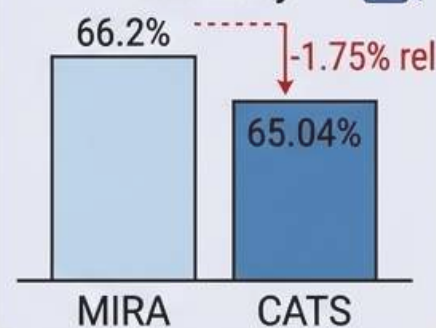
MIRA efficiency

(d) Detailed Performance Scores (Accuracy)

LongVideoBench (LVB)
Accuracy



VideoMME (V-MME)
Accuracy



CATS (ours) accuracy

MIRA accuracy